Physical Chemistry

Kinetics

(1) Consider the following mechanism used to describe the decomposition of a gaseous molecule

$$A + A \underset{k_{-2}}{\overset{k_2}{\leftrightarrow}} A^* + A$$

$$A^* \xrightarrow{k_1} \text{Products}$$

(A) Write the differential rate equation for $-dC_A/t$ and dC_{A*}/t and assuming a steady-state approximation for C_{A*}, write $-dC_A/t$ in terms of C_A and rate constants. (Show all of your mathematical analysis)

(B) Under what conditions is this a pseudo-first order reaction? Under what conditions is this a pseudo-second order reaction? (Show your answers in mathematical form and explain them).

(2) The second order reaction

$$OH^- + C_2H_5OH \text{ -----> } C_2H_50^- + H_2O$$

When studied in 0.1 M NaOH gave the pseudo-first-order rate equation

$$-d(C_2H_5OH)/dt = k(C_2H_5OH)$$

With $k=3x10^5\ s^{-1}$. What is the rate constant for the second-order reaction?